

Digital infra: Pushing chips hard

At the *Business Standard* Manthan on March 27, 2024, Ashwini Vaishnaw, minister of information technology, telecom and railways, made some hard-hitting points:

(i) Semiconductor is a \$750 billion global industry set to double in the next 6-7 years, and India is purposefully striving to develop its own capabilities.

(ii) What is working in favour of India is its abundant technically oriented talent, access to green energy and a growing specialty chemicals manufacturing ecosystem.

(iii) India is aiming to capture a significant share of the full value chain — from conceptualising a new chip, to designing and validating it, and then taking it to final fabrication.

(iv) India's moves in this regard are also aimed at bolstering its economic security.

His remarks came at a time when the action is indeed hotting up in the global marketplace.

The US Commerce Department is in the process of structuring a \$39 billion grants facility for chipmaking. It is understood that the USA will award grants totalling \$11.6 billion to Taiwan Semiconductor Manufacturing Company (TSMC) to assist the world's top chipmaker expand its manufacturing footprint in America. Other companies lining up for US grants are Intel, Secure Enclave, Global Foundries, Microchip Tech, BAE Systems, and a few others. Similar moves are being made by Japan, and the EU.

In India, the government has approved the establishment of semiconductor units as part of the "Development of Semiconductors and Display Manufacturing Ecosystems in India" policy, notified on December 21, 2021, with a total outlay of ₹76,000 crore. This policy aims to make India self-reliant, and to position it as a global hub for electronic system design and manufacturing. To attract investments in this field, the central government will provide financial support of 50 per cent of the project cost. Specifically, the policy will incentivise the setting up of semiconductor "fabs" (fabrication units), display fabs, and facilities for compound semiconductors, silicon photonics, sensors, and assembly and testing. In addition, state

governments are seen chipping in too. For example, beyond the 50 per cent central government support, the Gujarat government has also committed 20 per cent capital support to the Tata project.

Projects such as Tata's are already underway as part of this policy. Micron's assembly and testing facility in Gujarat's Sanand began construction in September 2023. Tata Electronics, partnering with Taiwan's Powerchip Semiconductor Manufacturing Corp (PSMC), is getting ready to set up a semiconductor fab in Gujarat's Dholera with an investment of ₹91,000 crore. The second project spearheaded by Tata Semiconductor Assembly and Test (TSAT) Pvt Ltd, will establish an assembly packaging and testing facility in Assam's Morigaon with an investment of ₹27,000 crore. Lastly, CG Power, in collaboration with Renesas Electronics Corporation and Stars Microelectronics, will set up an assembly and testing facility in Gujarat's Sanand with an investment of ₹7,600 crore. The Tata-PSMC proposal is the only "fab" that the government has approved as of now. It is also reliably learnt that some other world leaders are in active discussions with the government to finalise their entry strategies for India, including location and identification of local partners.

In the context of technology and electronics, "chips" and "semiconductors" are closely related but not the same. A semiconductor is a material that has electrical conductivity between that of a conductor and an insulator. A "chip", on the other hand, typically refers to an integrated circuit (IC) or microchip, which is a small semiconductor wafer on which thousands or millions of tiny electronic components like transistors, capacitors, and resistors are fabricated. These components are interconnected to perform a specific function, such as processing data in a computer or controlling electronic devices. Chips are highly complex products to manufacture. The critical stage of fabrication takes place in specialised facilities known as semiconductor fabrication plants or "fabs", where processes like photolithography, etching, doping, and metallisation are employed to create the chip's physical structure on silicon wafers.

This is a unique industry where the annual investment in R&D is often as much as capital invested. The need for deep technical knowhow and competitive costs have resulted in an extremely specialised global supply chain with very few players. Semiconductors provide the essential functionality for electronic devices to process, store and transmit data. They are integrated circuits made from silicon by packing billions of electronic components in a few square millimetres. Chips are critical to all electronics around us — from smartphones to servers, modern cars, machinery and automation, critical infrastructure and even defence systems.

Despite India's growing contribution to end-user demand (particularly driven by the booming smartphone market, which reached \$140 billion in FY22), the country remains a passive consumer, vulnerable to geopolitical tensions that could disrupt the supply chain. For instance, the Covid-19 chip shortage and US sanctions on Huawei highlighted India's dependency on external suppliers, exposing it to risks of supply chain disruptions. Currently, India imports 100 per cent of its semiconductors. Projections indicate that import figures could surge to \$110 billion by 2030, underlining the critical need to localise the semiconductor supply chain to align with the vision of a self-reliant India in this area of critical and high-tech electronics manufacturing.

The Government of India has clearly taken a series of decisive measures. Starting in 2019, it awarded "infrastructure status" to semiconductor fabrication facilities. This was the first time a manufacturing facility had been included in the official definition of infrastructure. The process of disbursing the capital subsidy marked the next logical step. Considering the criticality of this industry, the global, technical and investment challenges, the government might find it appropriate to increase the size of the Capital Subsidy Fund to cater to the interest being shown by potential entrants.

The country now looks forward to confidently powering its digital infrastructure with a few more committed players, signifying that the chips are not down anymore!

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